MILITARY SPECIFICATION

THERMISTOR, (THERMALLY SENSITIVE RESISTOR), INSULATED NEGATIVE TEMPERATURE COEFFICIENT STYLE RTH16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RTH16 thermistors. This style is available in resistance ratios A, B, and C. The only terminal lead available is S type. Resistance tolerance versus temperature characteristics F, G, J, and K are applicable through the maximum temperature of 125°C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-T-23648	-	Thermistor, (Thermally Sensitive Resistor) Insulated, General Specification For.

STANDARD

MILITARY

MS90178	-	Standard 24-Value Series Decade for Electronic Components of 5-, 10-, and 20- Percent Tolerances.

(Copies the of specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ES, 1507 Wilmington Pike, Dayton, OH 45444-5276, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

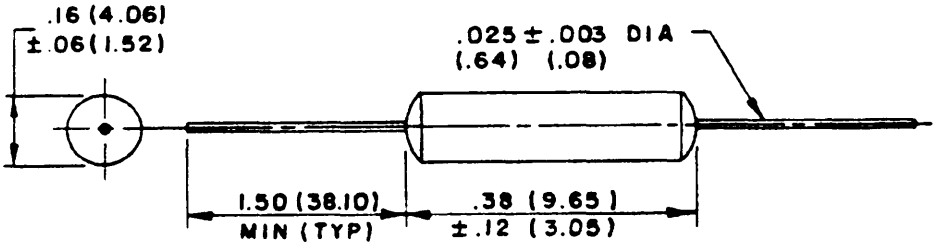
3.1 Requirements. Requirements shall be in accordance with MIL-T-23648, and as specified herein.

3.2 Design and construction. The thermistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Thermal time constant. The thermal time constant shall be 100 seconds maximum.

3.4 Dissipation constant. The dissipation constant shall be 2.5 milliwatts per °C minimum.

3.5 Power rating. The thermistor shall be capable of dissipating a maximum power of .25 watts at 25°C. Thermistors shall be derated in accordance with figure 2.



NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.

FIGURE 1. Style RTH16 thermistors.

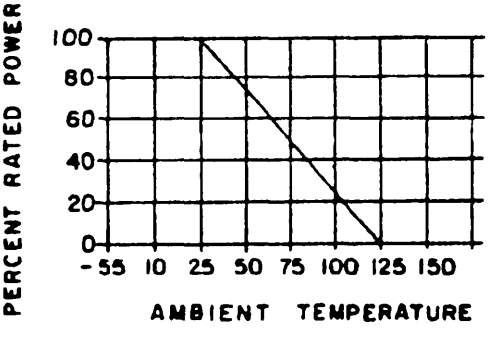


FIGURE 2. Derating curve for high ambient temperatures.

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3.6 Resistance.

3.6.1 Resistance at 25°C. Standard values shall be as specified in MS90178 except that the 5 percent decade values shall also be used for resistance tolerance F and G.

TABLE I. Minimum and maximum resistance values (at 25°C) for each available resistance ratio.

Ratio 19.8 (A)	Ratio 29.4 (B)	Ratio 48.7 (C)
ohms	ohms	ohms
390 min	2.7k min	56k min
1.5k max	12k max	120k max

3.6.2 Resistance at temperatures other than 25°C. Resistance at temperatures other than 25°C are determined by multiplying the zero-power resistance at 25°C by the factors shown in table II for the applicable resistance ratio.

TABLE II. Resistance temperature characteristic factors.

Temperature	Ratio 19.8 (A)	Ratio 29.4 (B)	Ratio 48.7 (C)
-55	54.9	100	---
-15	5.77	7.38	8.80
0	2.85	3.27	3.66
25	1.00	1.00	1.00
50	.405	.360	.320
75	.184	.148	.116
100	.0923	.0675	.047
125	.0503	.0340	.0205

3.7 Short time load. The maximum allowable change in zero-power resistance as the result of the short time load test shall be ±2 percent.

3.8 Low temperature storage. The maximum allowable change in zero-power resistance as the result of the low temperature storage test shall be ±2 percent.

3.9 High temperature storage. The maximum allowable change in zero-power resistance as the result of the high temperature storage test shall be ±1 percent.

3.10 Terminal strength. When tested in accordance with 4.3, the maximum allowable change in zero-power resistance shall be ±1 percent.

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3.11 Thermal shock. The maximum allowable change in zero-power resistance as the result of the thermal shock test shall be ±2 percent.

3.12 Resistance to soldering heat. The maximum allowable change in zero-power resistance as the result of the resistance to soldering heat test shall be ±1 percent.

3.13 Moisture resistance. The maximum allowable change in zero-power resistance as the result of the moisture resistance test shall be ±5 percent.

3.14 Life. The maximum allowable change in zero-power resistance as the result of the life test shall be ±5 percent.

3.15 High temperature exposure. The maximum allowable change in zero-power resistance as the result of the high temperature exposure test shall be ±1 percent after 100 hours, and ±2 percent after 1000 hours.

3.16 Vibration, high frequency. The maximum allowable change in zero-power resistance as the result of the vibration test shall be ±2 percent.

3.17 Shock, specified pulse. The maximum allowable change in zero-power resistance as the result of the shock test shall be ±2 percent.

3.18 Immersion. The maximum allowable change in zero-power resistance as the result of the immersion test shall be ±3 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification inspection. Qualification inspection is not applicable to this document.

4.2 Sampling and inspection. The sampling, acceptance inspection, and test procedures specified in tables IV and V of MIL-T-23648 are applicable as sampling, inspection, and test procedures of this detail specification.

4.3 Terminal strength.

4.3.1 Direct load. Direct load shall be applied gradually until the load reaches 4.5 pounds.

4.4 Solderability. The solderability test is applicable to this specification.

4.5 Resistance to soldering heat. The resistance to soldering heat is applicable to this specification.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-T-23648.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-T-23648 are applicable.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

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Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5905-1129-4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-23648/6A	THERMISTOR, (THERMALLY SENSITIVE RESISTOR), INSULATED NEGATIVE TEMPERATURE COEFFICIENT STYLE RTH16

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

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